

BeeWave: Creta swarm kinetic movement using by SMA display and embedded cellular automata-based mechanism

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ABSTRACT

"Bee Wave" is a kinetic art sculpture driven by a shaped memory alloy. It is driven by the material properties of the memory alloy, which is heated and contracted to pull the lever on the drive to pull the acrylic wings, making them sway like the wings of a bee in nature (Figure 1). Since no motor is used, it is quiet and elegant; the cell of the triangular cone is like a cellular automata, which has a mechanism to receive, store and process signals transmitted between cells and cells, and can influence the neighboring cells, from the individual triangular cone to influence the three neighboring cells, through the rule of triggering all the way to the group, like a wave cycle; and in the process of implementation into the nature of the existence of In the process of the actual work, the biological characteristics and appearance of the bionic design are incorporated, such as the adoption of a biological appearance, the mechanism of the bee to capture the hidden audio conversion behavior, the memory alloy of the shape like a living creature, and the function of the cellular automaton to transmit signals in the game of life [Conway 1970], and the above mechanisms are "mixed" to create this creation (Bee Wave).

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1 INTRODUCTION

In the field of architecture, most of them refer to the appearance and skeleton of natural creatures or plants to create a new architectural vocabulary or shape and apply it to the overall appearance of the building, which will be like a sculpture standing in nature. Theo Jansen, a famous artist in the field of art, made an installation bionic beast [Jansen 2020], which is a combination of art, bionic, engineering, and performance, achieving self-realization and driving of the installation through natural forces. (Bee Wave) combines the advantages of the preceding cases and conducts research on the bionic behavior between signal transmission, together with the

material of biological muscle (shape memory alloy) and the mechanism of audio-triggered movement, and a series of experiments to achieve the presentation of an organic life form from.

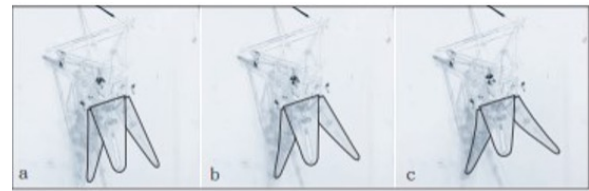


Figure 1: Explanation of the operation process of the wing drive (a) The alloy wire is not yet heated. (b) The alloy wire is heating up. (c) After the alloy wire is heated.

Number of Module	1 side	12 sides	40 sides
3D Render Model			
Physical Installation			

Figure 2: Combination of Bee Wave units.

2 BACKGROUND

Observe the current art of kinetic art, the impression is that through the combination of countless gears and motors to drive and complete a series of operations, mechanical sense is stronger, but the more time, the development process also becomes more limited .However, through case studies, we learned about the mode of operation of organisms in nature, and discovered the focus of research on group behavior. Moreover, the game of life is found in the 2D program, based on simple rules, it is possible to generate a coherent structure of cells, which will give rise to many diffuse forms and interesting group behaviors, and what will happen if the bionics between signal transmission happens on this installation (BeeWave). Through case studies in the field of art, the artist (Akira Nakayasu) [Nakayasu 2019] created a bionic device like a floating sea anemone, the beauty of the individual movement and the visual effect of the group movement; the artist (Philip Beasley) [Beasley 2009] designed the structure and movement of the installation, and understood how

the operation of the installation could influence the environment. By using memory alloy as the core, it is very secret and attractive in this creation. The transmission of signals between group behaviors is programmed inside the device. In the past, cellular automata was mostly an iterative relationship in 2D. , But this installation (BeeWave) explores what will happen in a 3D sphere (Figure 2)? It is like a collection of the operation patterns of some creatures in nature, such as the "hot bee wave" produced by the defense process of bees [Chen 2015], the cell's birth and death, the signal spreads out like waves in the process; it's like seabirds changing patterns in flight; the above focus combines the new biological action to occur on (BeeWave), through environmental information or The interaction of the viewers triggers the wing drive and gradually enlarges the group behavior into a sphere.

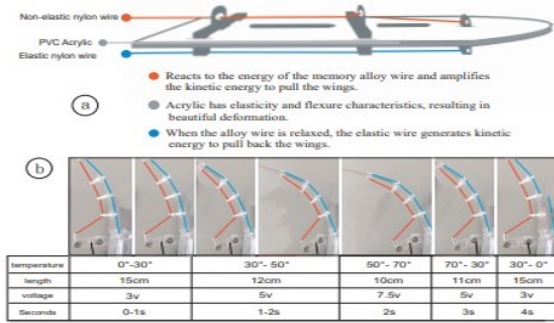


Figure 3: About the wing drive (a) The material and function of the wing are explained. (b) The process of pulling the wings by the alloy wire and the deformation and various values.

3 DESIGN METHOD

The external shape of the device is analyzed by observing the structure of living creatures and plants, firstly, the two-way stress of opening and closing is analyzed by taking Mimosa as an example (Figure 3) and applied to the bee's wings, from the bee's body and wings are simplified into each unit part, each unit part is cut by acrylic laser, firstly, the parts connected with the memory alloy wire are cut and assembled into a wing driver. There are three pieces of acrylic wings, kinetic levers, and a 26 cm, 0.02 mm thick shaped memory alloy wire through the center of the device, the elastic line is distributed behind the wings (Figure 4), the electronic components Arduino, through the power 7.5V output signal, and heating the shape of the memory alloy wire, when the temperature reaches more than 70 degrees, the alloy wire will contract and deformation to pull the lever to generate power, the power will indirectly When the temperature reaches above 70 degrees, the alloy wire will shrink and deform and pull the lever to generate power, which will indirectly pull the wings above and the bound elastic wire behind to generate a displacement similar to the memory alloy wire, and when the heating is over, the alloy wire will relax and become longer, then the elastic wire behind will generate kinetic energy similar to the alloy wire to pull the wings and lever back to the initial position.

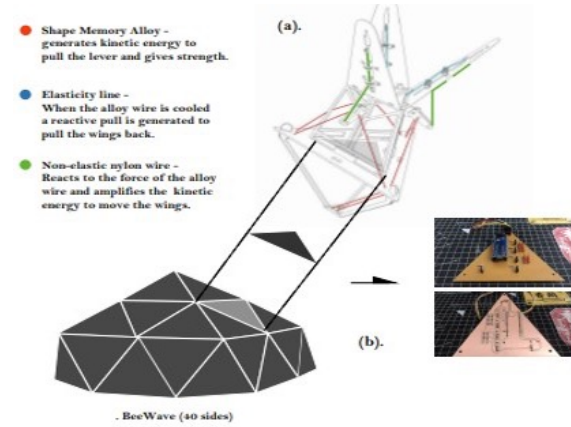


Figure 4: Explanation of the structure on the wing drive (a) Functional description of the alloy wire, elastic wire, non-elastic wire, and other parts. (b) The position of the custom circuit board in relation to the overall device.

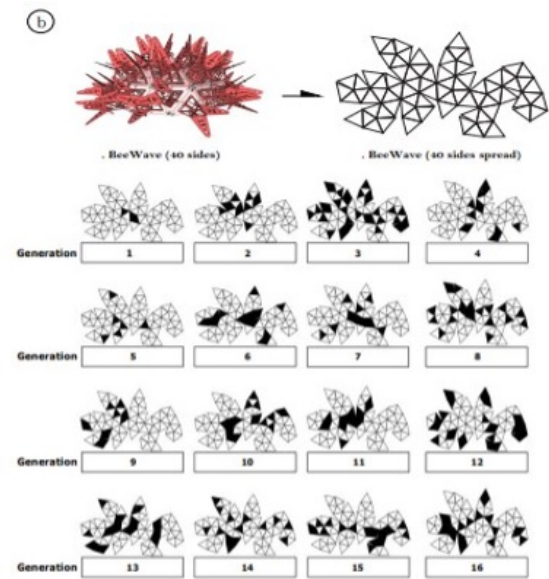


Figure 5: About the cellular automaton. (a) The rules of cell proliferation and death are explained. (b) The diffusion of cells on a forty-sided grid when the device is spread out.

4 CELLULAR AUTOMATA

The internal program is programmed by the cellular automata in the game of life, proposed by mathematician John Horton Conway, by spreading the device into a forty-sided triangular grid (Figure 5), each grid represents a cell, each cell has three neighbors, these cells have two states: (alive or dead), (open and closed), through the rules can be extended to countless life forms, the rules are like when there are two or three surviving neighboring cells, self-state death ; when there are no neighboring cells, self-state survival. Through

a series of cell survival and death processes, a biological behavior like a bee wave will be generated on the device, achieving a bionic design in programming.

5 FUTURE DEVELOPMENT

At present, the wing drive has developed to forty faces, and the design of the mechanism can be improved, such as whether the distribution nodes of the artificial wings can also find the appropriate rules from nature, or the materials can be used to visually enhance the materials that will change in response to environmental changes, and the programming of the cellular automaton will also be more diverse [Schiffman 2012], after which sensors will be added to sense environmental information, plus the audience's. Finally, after the BeeWave is modified and refined in detail, it will

be used as a large scale kinetic sculpture for art exhibitions or as an interesting interactive light fixture in the design field.

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